Report 6: Peak-to-Average Power Ratio of OTFS

EE451 — Zak Transform / Delay–Doppler Communication
Simulation S6

Mayank Mudgal (23B3918)

October 12, 2026

1 What this report is about

- A transmitter's amplifier must handle the *peaks* of the waveform, not just its average power. The ratio is the PAPR (peak-to-average power ratio); lower is better.
- The course slides (p. 15) say the Gaussian×sinc pulse is used "for reducing PAPR". Is that true? How does OTFS compare with OFDM?
- This report measures the PAPR of OTFS with three pulses (rectangular, Gauss×sinc, orthogonalised) and of OFDM, and looks at what the pulse shape *does* buy.

2 The concept in brief

- **Definition.** $\mathrm{PAPR} = 10\log_{10}\dfrac{\max_a |s[a]|^2}{\mathrm{mean}_a |s[a]|^2}$ in dB, measured here over one whole frame.
- **Many symbols add up like noise.** Every waveform sample is a sum of many independent random symbols, so it is nearly Gaussian. A Gaussian-like signal has occasional large peaks: for about 2000 independent Gaussian samples the PAPR is around 9 dB. Expect OTFS and OFDM to land near this.
- **One DD point is a very peaky signal.** A single DD symbol is a pulse train with one pulse per Zak period (Report 2): the signal is "on" for only $1/M$ of the time, so $\mathrm{PAPR} = M$, i.e. $10\log_{10} M$ dB. Filling more DD points mixes more terms and the PAPR falls.
- $N = 1$ **is single-carrier.** With one Doppler bin the IZak does nothing: the signal is just the symbols one after another, and its PAPR is that of the constellation plus the pulse.
- **Pulse shape.** Gauss×sinc and the orthogonalised pulse are band-limited; the rectangular pulse is not. The pulse's main benefit should therefore be a *compact spectrum*. Whether it also lowers the PAPR is what we test.

One-line memory aid: many random symbols ⇒ Gaussian-like signal ⇒ PAPR ≈ 8–9 dB, almost whatever the transform. A sparse DD grid is peakier ($10\log_{10} M$ for one point); pulse shaping buys spectrum, not PAPR.

3 What was simulated

- Grid M=32, N=16, D=4 (2048 samples per frame), random 16QAM symbols, 1000 frames; pulses with 65 taps.
- Four waveforms: OTFS with rectangular, Gauss×sinc ($\alpha = 4$) and orthogonalised ($\alpha = 16$ base pulse) pulses, and OFDM with M=32 subcarriers and N=16 symbols (same samples per frame).

- PAPR statistics, the effect of the number of occupied DD points K, the effect of N, and the power spectrum of the three OTFS waveforms.

4 Results and discussion

4.1 What the waveforms look like



Figure 1: Instantaneous power relative to the mean for one random frame. Dashed red line: the PAPR of that frame.

- **Expected:** all three look like noise with occasional peaks.
- **Observed:** yes. The rectangular waveform is a staircase (each value is held for D samples); the other two are smoother. For this one frame the PAPRs are 9.30 dB (rect), 9.31 dB (orthogonalised) and 8.38 dB (OFDM).
- One frame is only one random draw, so the statistics below are needed before drawing conclusions.

4.2 PAPR statistics over 1000 frames

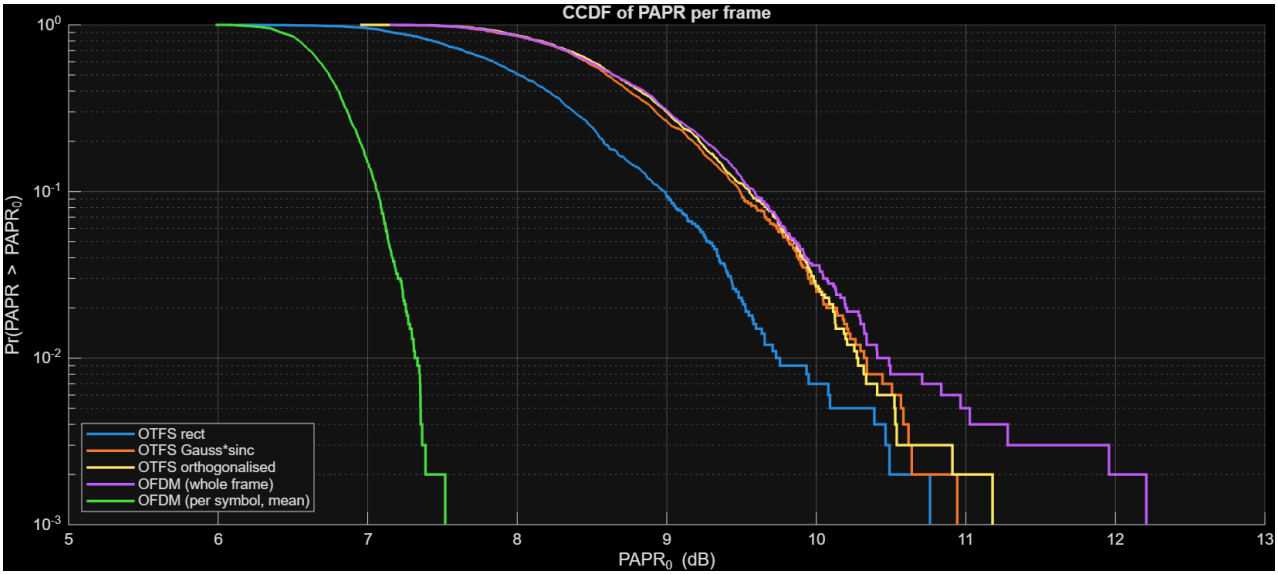


Figure 2: CCDF of the PAPR per frame: the probability that the PAPR exceeds the value on the horizontal axis.

Table 1: PAPR over the 1000 frames (dB).

Waveform	mean	99 %	99.9 %	max
OTFS, rectangular	8.04	9.71	10.76	10.76
OTFS, Gauss×sinc	8.65	10.30	10.94	10.94
OTFS, orthogonalised	8.70	10.25	11.18	11.18
OFDM, whole frame	8.71	10.40	12.21	12.21
OFDM, per symbol (mean over the 16 symbols)	6.74	7.31	7.52	7.52

- **Expected:** all frame-level PAPRs close to the Gaussian value of roughly 8–9 dB.
- **Observed:** the three band-limited waveforms (two OTFS pulses and OFDM) have almost the same CCDF and nearly the same mean (8.65–8.71 dB). Rect OTFS is lower by about 0.6 dB on average, and its 99% value is 9.7 dB against 10.3–10.4 dB for the others. OFDM has the longest tail (max 12.2 dB against 10.8–11.2 dB for OTFS in these 1000 frames).
- **OFDM per symbol** is lower (6.74 dB) simply because it is measured over 32×4 samples instead of the whole frame; a longer observation window tends to see higher peaks. The fair comparison with OTFS is the whole-frame OFDM value.
- Takeaway: OTFS is neither better nor worse than OFDM in PAPR at the frame level; the slides' claim that the Gauss×sinc pulse *reduces* PAPR is not supported here (it is about 0.6 dB *higher* than rect).

4.3 A single DD point is peaky

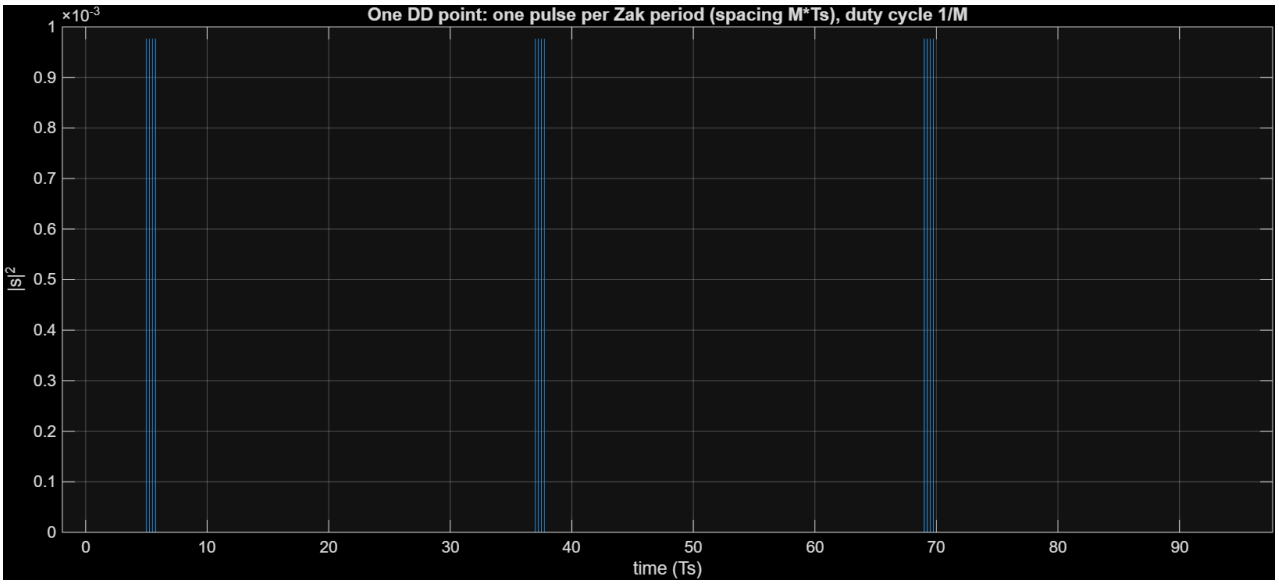


Figure 3: $|s|^2$ for one occupied DD point (rectangular pulse): one short burst per Zak period, $MT_s = 32\,T_s$ apart.

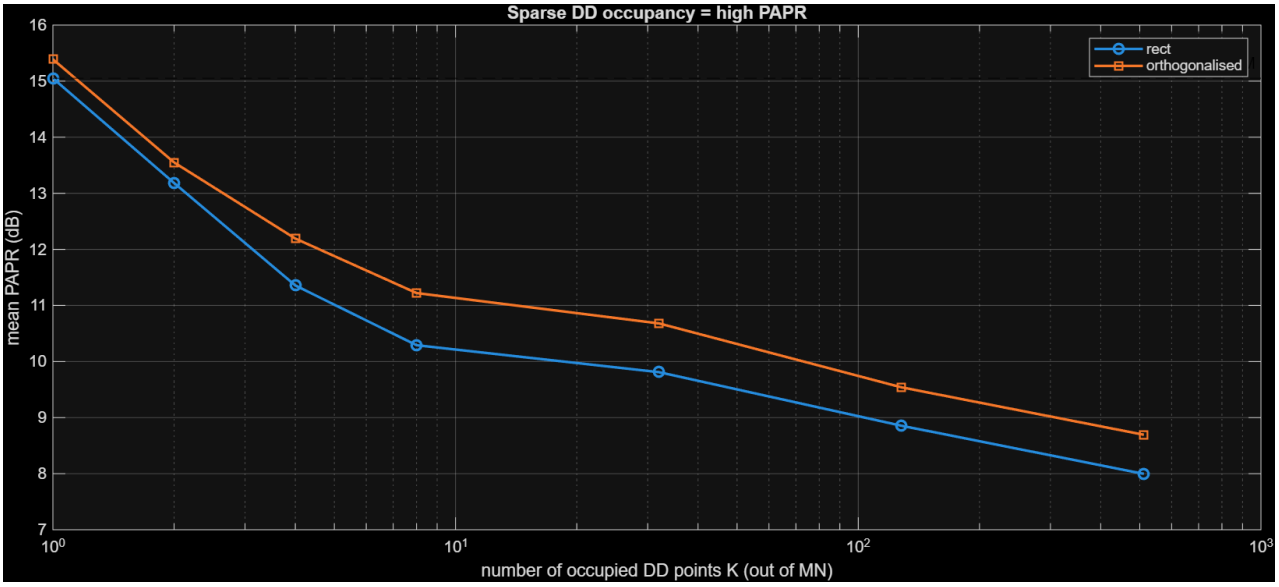


Figure 4: Mean PAPR against the number K of occupied DD points (out of $MN = 512$). Gray line: $10\log_{10} M = 15.05\,\text{dB}$.

- **Expected:** one point gives $10\log_{10} 32 = 15.05\,\text{dB}$; the PAPR falls as K grows.
- **Observed:** the burst occupies 1/32 of each period, and the PAPR with K=1 is 15.05 dB for rect (15.18 dB for Gauss×sinc and 15.39 dB for the orthogonalised pulse). It then falls steadily, to 8.00 dB (rect) and 8.69 dB (orthogonalised) with the full grid (K=512), close to the full-frame means of the table above.
- Takeaway: the PAPR is a property of how *full* the DD grid is. A lightly loaded frame, has a much higher PAPR; the same would be expected for a frame dominated by a few strong symbols such as pilots (not simulated).

4.4 Effect of the number of Doppler bins N

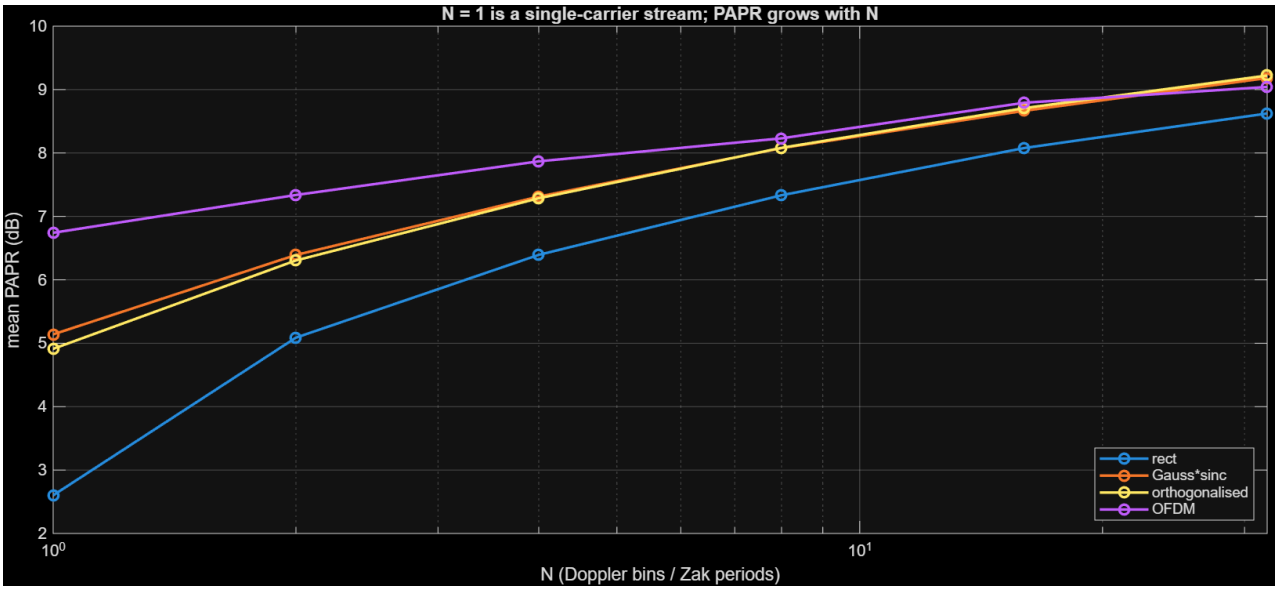


Figure 5: Mean PAPR against N (M=32, D=4, 200 frames per point).

- **Expected:** $N = 1$ is a plain single-carrier stream, and mixing more periods raises the PAPR towards the Gaussian level.
- **Observed:** for N=1 the rectangular pulse gives 2.60 dB, which is just the peak-to-mean ratio of 16QAM ($10 \log_{10} 1.8 = 2.55$ dB). The band-limited pulses give 5.14 dB (Gauss×sinc) and 4.91 dB (orthogonalised) (presumably the pulse's own ripple), and OFDM 6.74 dB. All curves rise with N: at N=32 rect is 8.62 dB and the other three are 9.04–9.22 dB. At N=16 the shaped OTFS pulses (8.67, 8.71 dB) and OFDM (8.79 dB) are about equal.
- Takeaway: OTFS with small N is closer to single-carrier and has a lower PAPR; larger N means more mixing, and a longer frame, so a higher PAPR.

4.5 What the pulse does buy: the spectrum

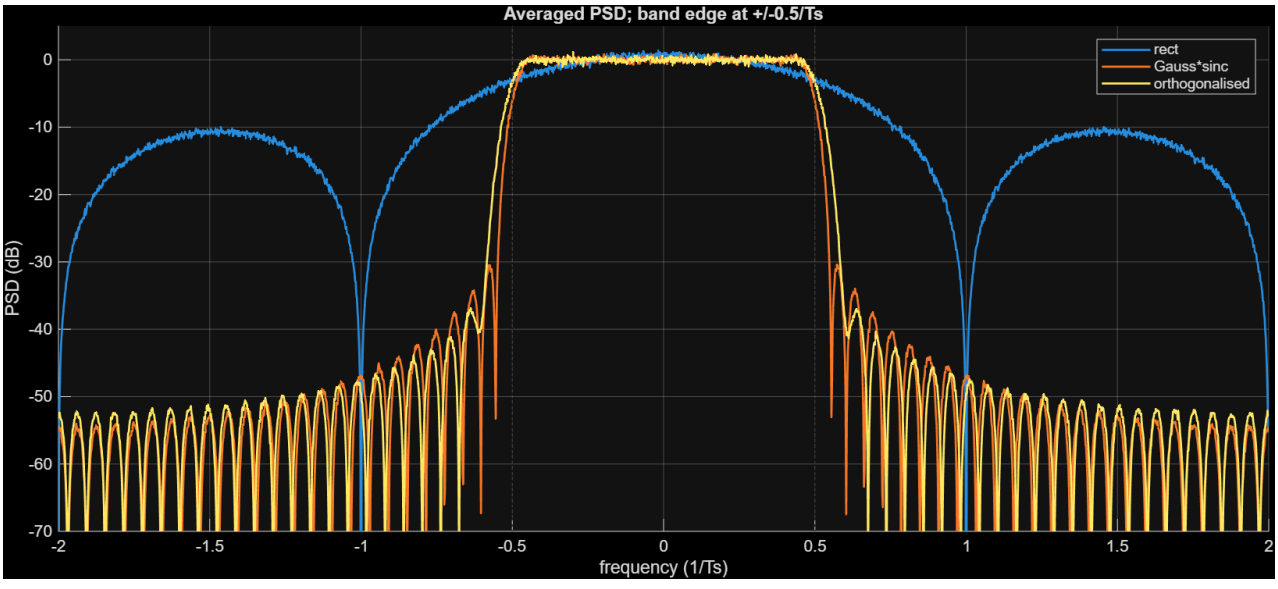


Figure 6: Averaged power spectral density of the OTFS waveforms (200 frames). Dotted lines: the nominal band edge $\pm 0.5/T_s$.

- **Expected:** the rectangular pulse leaks a lot of power outside the band (sinc-shaped sidelobes); the band-limited pulses should confine it.
- **Observed:** the rect spectrum has lobes peaking near $-10\,\mathrm{dB}$ around $\pm 1.5/T_s$, with nulls at $\pm 1/T_s$. Gauss×sinc and the orthogonalised pulse are flat inside the band and fall to about -30 to $-40\,\mathrm{dB}$ just outside it; further out they sit at about $-50\,\mathrm{dB}$ (set by cutting the pulse at 65 taps). The printed levels are $-9.4\,\mathrm{dB}$ (rect), $-40.7\,\mathrm{dB}$ (Gauss×sinc) and $-46.7\,\mathrm{dB}$ (orthogonalised) at $0.75/T_s$, and -10.8, -52.9 and $-51.8\,\mathrm{dB}$ at $1.5/T_s$: the shaped pulses are 31–37 dB lower at $0.75/T_s$ and about 41–42 dB lower at $1.5/T_s$.
- Takeaway: the shaped pulses trade a small increase in PAPR (0.6 dB) for a spectrum 30–40 dB cleaner out of band. That, not PAPR, is their real advantage.

5 Summary

- With a full DD grid, OTFS has a frame-level PAPR of about 8–9 dB, the same as OFDM measured over the same frame (8.7 dB).
- The rectangular pulse is about 0.6 dB lower, but its spectrum leaks 30–40 dB more power outside the band.
- The Gauss×sinc pulse does *not* reduce the PAPR in this simulation; its benefit is the compact spectrum.
- A sparse DD grid is peaky: one point gives $10\log_{10} M = 15\,\mathrm{dB}$. PAPR rises with the number of Doppler bins N; $N = 1$ is a single-carrier signal.

Appendix: details and assumptions

- PAPR is measured on the 4× oversampled digital waveform (peak sample over mean sample power, over one frame), with no clipping, no amplifier model and no reconstruction filter. The true analogue peaks would be slightly higher.
- OFDM here has M=32 subcarriers on the lowest 32 bins of a 4× zero-padded IFFT, N=16 symbols, no cyclic prefix, and the same 16QAM symbols as OTFS. The script does not compute an OFDM spectrum, so Fig. 6 shows OTFS only.
- The Gaussian reference value ($\approx 9.1\,\mathrm{dB}$ for 2048 independent complex Gaussian samples) comes from an independent NumPy simulation; the measured 8.7 dB is a little lower because oversampled samples are correlated.
- The 16QAM here uses levels $\pm 1, \pm 3$ (normalised to unit average power) without Gray mapping, which does not affect PAPR.
- The rectangular pulse is held for D samples and is not band-limited, so its PAPR and spectrum are not directly comparable with the band-limited pulses; the comparison shows the trade-off, not a like-for-like contest.
- All PAPR, fill-sweep, N-sweep and spectrum numbers are the values printed by the script (random seeds fixed). The 99% and 99.9% values rest on 1000 frames, so the latter, and the maximum, are only rough.
- The statement that the slide's Gauss×sinc claim is unsupported refers to this setup (pulse widths assumed as in Report 4); the slides do not say what it is compared with.
- The figures come from `S6_papr.m` (MATLAB dark theme). The histogram figure of the script is omitted, since the CCDF shows the same information.